Ensuring Reliable and Replicable Data Acquisition for Devices Under Test at the 3nm Bulk FinFET Node

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Abstract

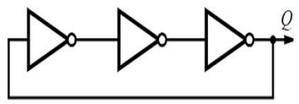
At advanced semiconductor technology nodes, manufacturing variabilities significantly impact the reliability of integrated circuits (IC's). Ring oscillator (RO) circuits are routinely used to measure variability, but are challenging to characterize directly and require frequency division. Such an approach reduces parameter resolution, requiring precise measurements. Establishing a reliable test setup to accurately capture small changes in RO parameter variations is crucial. In this work, a test setup is developed to guarantee cable connections to ensure reliable data acquisition. Test setup is used to measure RO frequencies at the 3nm bulk FinFET node to show variability in measured parameter.

Background

- Transistor scaling remains crucial for meeting the challenges of computation requirements
- ICs fabricated at advanced technology nodes face significant reliability challenges due to manufacturing variabilities
- RO circuits are used to measure variabilities. Variations in individual transistor parameters cause variations in RO frequency
- Measurement of RO frequency requires a custom test setup to ensure accuracy

Circuit and Test Setup Design

- Custom-designed test ICs containing ring oscillators fabricated in a commercial 3-nm bulk FinFET technology



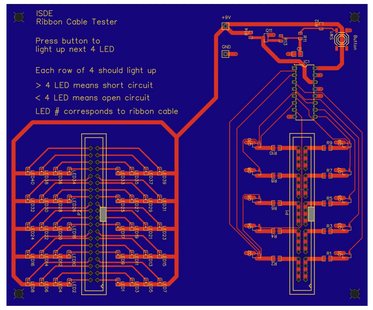
n-stage ring oscillator circuit diagram

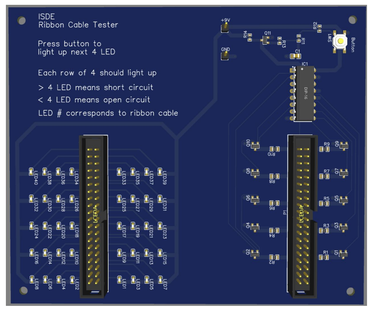
$$f = \frac{1}{2tn}$$

Gate delay of RO
t = time delay single inverter
n = number of inverters

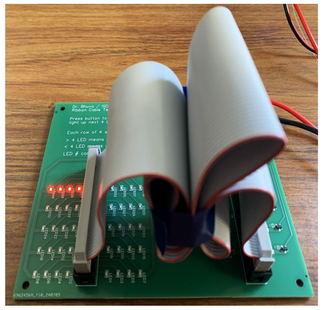
- Ring oscillator frequency measurement requires multiple inputs to be controlled and observed simultaneously
- Ribbon cables are best suited for such a test setup
- Ribbon cables are prone to damage if not handled carefully
- A ribbon-cable tester was developed to identify damaged ribbon cables to avoid downtime or faulty measurements
- Ribbon cable tester enables rapid fault detection
- Designed with EasyEDA using off-the-shelf components and a custom PCB for cost reduction
- Cable tester was designed for operation with a battery or a power supply
- Multiple iterations were designed and manufactured to optimize the final design
- The final printed circuit board design cost of $24 is significantly less than market average for similar devices
- The final design was thoroughly tested to verify proper operation

Ribbon Cable Tester PCB Design





Final design of Ribbon cable tester PCB



RO Frequency Measurements

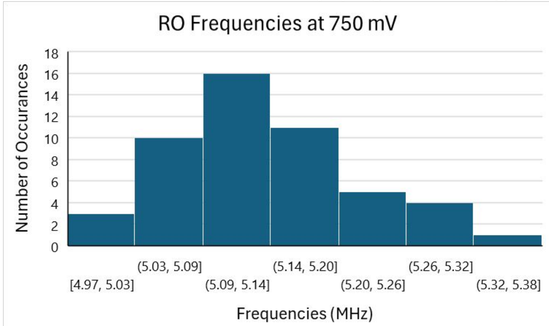
- Frequency was measured using MDO34 oscilloscope capable of measuring frequencies automatically
- RO frequencies were measured for 52 ICs, providing statistically significant results
- Frequencies were measured as a function of supply voltage
- In addition to nominal supply voltage (750 mV), frequencies were measured for 650 mV and 550 mV

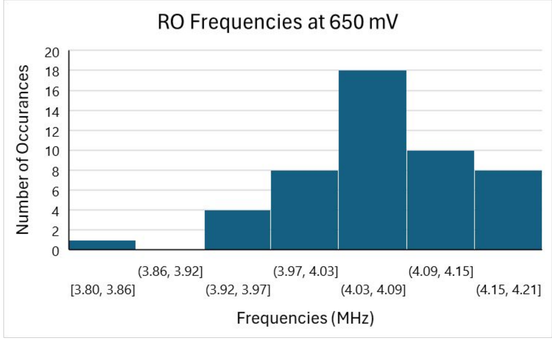
Results

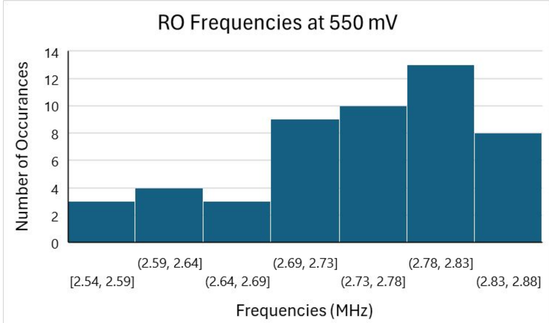
Mean and standard deviations for measurements

Voltage	Mean RO Frequency (MHz)	Std. Dev.
750	5.14	0.088
650	4.07	0.076
550	2.75	0.087

Histograms for RO frequencies at three different supply voltages







Conclusions

- Transistor scaling to the 3nm node requires robust testing methodologies to evaluate manufacturing variability
- Understanding variability in manufacturing processes lead to more reliable IC's
- Custom design for a ribbon cable tester allows for a robust and accurate test setup
- Variability in RO frequencies were very small (standard deviation was 1.7% of mean at nominal supply voltage)